

## CHAPTER III.6, WEEK 1

Noah Parker

6.1. Let  $(X, \mathcal{O}_X)$  be a ringed space, and let  $\mathcal{F}', \mathcal{F}'' \in \mathfrak{Mod}(X)$ . An extension of  $\mathcal{F}''$  by  $\mathcal{F}'$  is a short exact sequence

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

in  $\mathfrak{Mod}(X)$ . Two extensions are *isomorphic* if there is an isomorphism of the short exact sequences, inducing the identity maps on  $\mathcal{F}'$  and  $\mathcal{F}''$ . Given an extension as above, consider the long exact sequence arising from  $\text{Hom}(\mathcal{F}'', \cdot)$ , in particular the map

$$\delta : \text{Hom}(\mathcal{F}'', \mathcal{F}'') \rightarrow \text{Ext}^1(\mathcal{F}'', \mathcal{F}'),$$

and let  $\xi \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$  be  $\delta(1_{\mathcal{F}''})$ . Show that this process gives a one-to-one correspondence between isomorphism classes of extensions of  $\mathcal{F}''$  by  $\mathcal{F}'$ , and elements of the group  $\text{Ext}^1(\mathcal{F}'', \mathcal{F}')$ . For more details, see, e.g., Hilton and Stammbach, Ch. III.

*Proof.* We define a map from  $\text{Ext}^1(\mathcal{F}'', \mathcal{F}')$  to extensions  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{E} \rightarrow \mathcal{F}'' \rightarrow 0$  as follows. Fix an injective resolution  $0 \rightarrow \mathcal{F}' \xrightarrow{d^{-1}} \mathcal{I}^\bullet$  of  $\mathcal{F}'$  with maps  $d^i : \mathcal{I}^i \rightarrow \mathcal{I}^{i+1}$ , so that

$$\begin{aligned} \text{Ext}^1(\mathcal{F}'', \mathcal{F}') &\cong h^1(\text{Hom}(\mathcal{F}'', \mathcal{I}^\bullet)) \\ &\cong \frac{\ker d^1}{\text{im } d^0}. \end{aligned}$$

Thus, we can regard an element  $[\xi] \in \text{Ext}^1(\mathcal{F}'', \mathcal{F}')$  as the class of a map  $\xi : \mathcal{F}'' \rightarrow \mathcal{I}^1$  with  $d^1 \xi = 0$ . Fix such an element  $[\xi]$  with representative  $\xi$ , and take the pullout  $\mathcal{P} := \mathcal{I}^0 \times_{\mathcal{I}^1} \mathcal{F}''$  in the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & & & & \uparrow \gamma & \lrcorner & \uparrow \xi & & \\ & & & & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \end{array}$$

Then we have  $d^0 \phi = 0$ , so the universal property of  $\mathcal{P}$  supplies a unique map  $\phi : \mathcal{F}' \rightarrow \mathcal{P}$  such that the following diagram commutes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{d^{-1}} & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & & \nearrow d^{-1} & & \uparrow \gamma & \lrcorner & \uparrow \xi & & \\ & & \mathcal{F}' & \xrightarrow{\exists! \phi} & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \\ & & \searrow & & \downarrow & & \downarrow & & \\ & & & & 0 & & & & \end{array}$$

We will now show that  $0 \rightarrow \mathcal{F}' \xrightarrow{\phi} \mathcal{P} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$  is a short exact sequence.

$\phi$  is monic since  $d^{-1} = \gamma\phi$  is monic, and  $\psi\phi = 0$  by construction. To show exactness at the middle, it suffices to show that  $\text{im } \phi$  satisfies the universal property of  $\ker \psi$ . Let  $\alpha : \mathcal{A} \rightarrow \mathcal{P}$  be a morphism such that  $\psi\alpha = 0$ . It suffices to show that  $\alpha$  factors through  $\text{im } \phi$ . Recall that  $\phi, d^{-1}$  are monic and  $\mathcal{F}' \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1$  is exact, so

$$\text{im } \phi \cong \text{coim } \phi \cong \mathcal{F}' \cong \text{im } d^{-1} \cong \ker d^0.$$

Then  $d^0\gamma\alpha = \xi\psi\alpha = 0$ , so the universal property of kernels supplies a unique map  $\tilde{\alpha} : \mathcal{A} \rightarrow \mathcal{F}'$  such that  $\gamma\alpha = d^{-1}\tilde{\alpha}$ . Further,  $\gamma\phi\tilde{\alpha} = \gamma\alpha$  and  $\psi\phi\tilde{\alpha} = 0 = \psi\alpha$  imply that  $\alpha = \phi\tilde{\alpha}$  via universal property of pullbacks.

$$\begin{array}{ccccc} & & \mathcal{I}^0 & \xrightarrow{d^0} & \mathcal{I}^1 & \xrightarrow{d^1} & \mathcal{I}^2 \\ & \nearrow d^{-1} & \uparrow \gamma & \lrcorner & \uparrow \xi & & \\ \mathcal{F}' & \xrightarrow{\phi} & \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & & \\ & \nwarrow \exists! \tilde{\alpha} & \uparrow \alpha & \nearrow 0 & & & \\ & & \mathcal{A} & & & & \end{array}$$

Hence, the above diagram commutes, and  $\tilde{\alpha}$  is indeed unique since  $\phi$  is monic.

Finally, we show that  $\psi$  is epic.  $d^1\xi = 0$  allows us to factor  $\xi$  through  $\ker d^1 = \text{im } d^0$ , giving us the following diagram, where the rightmost square is a pullback.

$$\begin{array}{ccccc} \mathcal{I}^0 & \xrightarrow{\tilde{d}^0} & \text{im } d^1 & \xrightarrow{i} & \mathcal{I}^1 \\ \gamma \uparrow & & \uparrow \hat{\xi} & & \uparrow \xi \\ \mathcal{P} & \xrightarrow{\psi} & \mathcal{F}'' & \xrightarrow{\text{id}} & \mathcal{F}'' \end{array}$$

It follows from the following elementary lemma on pullbacks that the lefthand square is a pullback as well.

**Lemma 1.** *Suppose that the rightmost square is a pullback. Then the leftmost square is a pullback if and only if the entire rectangle is a pullback.*

$$\begin{array}{ccccc} \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \\ \uparrow & & \uparrow & \lrcorner & \uparrow \\ \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \end{array}$$

Recall that coequalizers are epic in any category. Conversely, an epic morphism in an abelian category is the cokernel of its kernel, so, in particular a coequalizer. Thus, we say that abelian categories are *regular*.  $\psi$  epic will follow from the following result on general categories.

**Proposition 1.** *Let  $\mathcal{A}$  be an abelian category. We say that an epimorphism  $f : A \rightarrow B$  is regular if it is the coequalizer of some pair of morphisms  $c_1, c_2 : C \rightarrow A$ . Then*

- (a)  $\mathcal{A}$  is finitely complete;

- (b) If  $f : A \rightarrow B$  is a regular epimorphism and  $B' \rightarrow B$  is a morphism, then the pullback  $f' : A' \rightarrow B'$  depicted below is a regular epimorphism;

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \uparrow & \lrcorner & \uparrow \\ A' & \xrightarrow{f'} & B' \end{array}$$

- (c) Every epimorphism is regular.

We say that  $\mathcal{A}$  is a regular category. In particular, epimorphisms in an abelian category are preserved under pullbacks.

*Proof.* (a) holds by definition, as does (c) after noting that coequalizers and cokernels coincide in an abelian category. Thus, it suffices to show (b). Namely, that the pullback of a cokernel is a cokernel.

Suppose that  $f : A \rightarrow B$  is the cokernel of  $g : C \rightarrow A$ . Then there is a unique map  $g' : C \rightarrow A'$  such that the following diagram commutes.

$$\begin{array}{ccccc} & & A & \xrightarrow{f} & B \\ & \nearrow g & \uparrow a & \lrcorner & \uparrow b \\ & & A' & \xrightarrow{f'} & B' \\ & \nwarrow \exists! g' & & & \\ C & & & & 0 \end{array}$$

Taking the cokernel  $f'' : A' \rightarrow B''$  of our new map  $g' : C \rightarrow A'$  gives us a unique map  $b' : B'' \rightarrow B'$  such that the following diagram commutes.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \uparrow a & \lrcorner & \uparrow b \\ A' & \xrightarrow{f'} & B' \\ \uparrow \text{id} & & \uparrow \exists! b' \\ A' & \xrightarrow{f''} & B'' \end{array}$$

□

□

6.2. Let  $X = \mathbb{P}_k^1$ , with  $k$  an infinite field.

- (a) Show that there does not exist a projective object  $\mathcal{P} \in \mathfrak{Mod}(X)$ , together with a surjective map  $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow 0$ . [*Hint:* Consider surjections of the form  $\mathcal{O}_V \rightarrow k(x) \rightarrow 0$ , where  $x \in X$  is a closed point,  $V$  is an open neighborhood of  $x$ , and  $\mathcal{O}_V = j_!(\mathcal{O}_X|_V)$ , where  $j : V \rightarrow X$  is the inclusion.]
- (b) Show that there does not exist a projective object  $\mathcal{P}$  in either  $\mathfrak{QCoh}(X)$  or  $\mathfrak{Coh}(X)$  together with a surjection  $\mathcal{P} \rightarrow \mathcal{O}_X \rightarrow 0$  [*Hint:* Consider surjections of the form  $\mathcal{L} \rightarrow \mathcal{L} \otimes k(x) \rightarrow 0$ , where  $x \in X$  is a closed point, and  $\mathcal{L}$  is an invertible sheaf on  $X$ .]

*Proof.* Take a closed point  $x \in X$ , and let  $U, V \ni x$  be neighborhoods of  $x$  such that  $U \not\subset V$  and  $V \not\subset U$ .  $\square$

6.3. Let  $X$  be a noetherian, and let  $\mathcal{F}, \mathcal{G} \in \mathfrak{Mod}(X)$ .

- (a) If  $\mathcal{F}, \mathcal{G}$  are both coherent, then  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  is coherent, for all  $i \geq 0$ .
- (b) If  $\mathcal{F}$  is coherent and  $\mathcal{G}$  is quasi-coherent, then  $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$  is quasi-coherent, for all  $i \geq 0$ .

*Proof.* (a) We begin by showing the result for  $i = 0$ . Take an affine open  $\text{Spec } A \cong U \subset X$  and finitely generated  $A$ -modules  $M, N$  such that  $\mathcal{F}|_U \cong \widetilde{M}$  and  $\mathcal{G}|_U \cong \widetilde{N}$ . Then

$$\text{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U) \cong \text{Hom}_A(M, N)$$

is a finitely generated  $A$ -module. Namely, it's generated by  $\psi_{m,n} : M \rightarrow N : x \mapsto \delta_{x,m}n$ . **(Is this really true?)** In particular, we can take a distinguished affine  $D(f) \subset \text{Spec } A$  for  $f \in A$ , and see that

$$\begin{aligned} \text{Hom}_{\mathcal{O}_{D(f)}}(\mathcal{F}|_{D(f)}, \mathcal{G}|_{D(f)}) &\cong \text{Hom}_{A_f}(M_f, N_f) \\ &\cong \text{Hom}_A(M, N)_f, \quad \textbf{(Please justify this!)} \end{aligned}$$

whence

$$\mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U) \cong \widetilde{\text{Hom}_A(M, N)}.$$

Now, fix an open affine  $\text{Spec } A \cong U \subset X$  once more so that  $\mathcal{F}|_U = M$  and  $\mathcal{G}|_U = N$ , and take a free resolution  $L_\bullet \rightarrow M \rightarrow 0$ , with  $L_i = A^{l_i}$  for  $l_i \geq 0$ . Then

$$\mathcal{E}xt$$

$\square$

6.5. Let  $X$  be a noetherian scheme, and assume that  $\mathfrak{Coh}(X)$  has enough locally frees (Ex. 6.4). Then for any coherent sheaf  $\mathcal{F}$  we define the *homological dimension* of  $\mathcal{F}$ , denoted  $\text{hd}(\mathcal{F})$ , to be the least length of a locally free resolution of  $\mathcal{F}$  (or  $+\infty$  if there is no finite one). Show:

- (a)  $\mathcal{F}$  is locally free  $\iff \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$  for all  $\mathcal{G} \in \mathfrak{Mod}(X)$ ;
- (b)  $\text{hd}(\mathcal{F}) \leq n \iff \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) = 0$  for all  $i > n$  and all  $\mathcal{G} \in \mathfrak{Mod}(X)$ ;
- (c)  $\text{hd}(\mathcal{F}) = \sup_x \text{pd}_{\mathcal{O}_x} \mathcal{F}_x$ .

6.8.